DEEPRESEARCH AGENT · RESEARCH REPORT

An Analysis of India's UPI Payment System and Its Growth

Topic: How does India's UPI payment system work, and how fast has it grown?

Research questions	6
Searches run	45
Sources read	15 from 10 websites
Claims extracted	65
Claim groups	Verified 3 · Likely 15 · Contested 1 · Unverified 15
Depth	standard
Generated	10 October 2026, 20:45

This report was written by an AI research agent. Every finding is linked to its sources and has a confidence label that estimates how well independent, trusted sources support it. The labels are not proof: check important facts in the cited sources before relying on them.

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1. Executive summary

- **VERIFIED** The Unified Payments Interface (UPI) was launched on 11 April 2016 by the National Payments Corporation of India (NPCI) under the regulatory oversight of the Reserve Bank of India (RBI) [1][2][3].
- **VERIFIED** UPI is a real-time, account-to-account payment system that enables instant, secure transactions via mobile apps [4][5].
- **VERIFIED** UPI has contributed to increased financial inclusion, reduced cash usage, and faster payment processing in India [2][6][7][3].
- **LIKELY** Evidence suggests that UPI consolidates multiple bank accounts into a single mobile interface [8].
- **LIKELY** Evidence suggests that UPI transaction volumes grew from 0.02 billion in 2017 to over 83 billion in 2024 [9][10].
- **LIKELY** Evidence suggests that the implementation of the Zero Merchant Discount Rate (MDR) in 2019 eliminated transaction fees for merchants to prioritize adoption velocity [10].

2. Research questions and method

- **Q1** What is India's Unified Payments Interface (UPI) and how does its foundational architecture work?
- **Q2** What are the key transaction statistics, volume numbers, and market share metrics for UPI in recent years?
- **Q3** What are the primary economic and technological causes behind the rapid adoption of UPI across India?
- **Q4** What effects has the widespread implementation of UPI had on traditional banking, cash usage, and financial inclusion in India?
- **Q5** What are the main disagreements, security concerns, or debates surrounding UPI transaction failures and market concentration risks?
- **Q6** What recent developments and international expansions have occurred for India's UPI payment system?

The Researcher agent ran 45 searches on DuckDuckGo, Wikipedia and OpenAlex, opened 15 sources from 10 different websites and gave each a trust tier. 5 pages could not be opened. The Extractor pulled 65 single-fact claims from the sources and kept a claim only if its supporting quote was found in the source text. The Verifier grouped claims that say the same thing (embeddings from gemini-embedding-2, cosine similarity ≥ 0.85), asked Gemini whether each source supports or contradicts each group, counted independent websites (copies of the same story count once) and gave every group a confidence label.

Confidence labels

Label	Rule used by the Verifier
VERIFIED	2 or more independent websites agree, at least one is tier 1 or 2, and no tier 1 or 2 source disagrees
LIKELY	Supported by one tier 1 source, or by 2 or more independent lower-tier websites
CONTESTED	A tier 1 or 2 source disagrees with the claim; both sides are shown in section 4
UNVERIFIED	Only one low-trust source; a targeted search was tried, and the claim is left out of the findings

Trust tiers

Tier	Weight	Type of website
1	0.9	Government, university, international organisation, scholarly journal
2	0.7	Reputable news agency or reference work (Wikipedia 0.6, arXiv preprints 0.75)
3	0.4	Any other website
4	0.1	Forum, social media or user-generated content (never above 0.2)

3. Findings

3.1 What is India's Unified Payments Interface (UPI) and how does its foundational architecture work?

The Unified Payments Interface (UPI) is a real-time, account-to-account payment system that enables instant, secure transactions via mobile apps [4][5]. It was launched on 11 April 2016 by the National Payments Corporation of India (NPCI) under the regulatory oversight of the Reserve Bank of India (RBI) [1][2][3].

Evidence suggests that the foundational architecture of UPI consolidates multiple bank accounts into a single mobile interface [8]. By bringing these different accounts together, the system facilitates the real-time, instant, and secure transactions characteristic of the platform [4][5].

3.2 What are the key transaction statistics, volume numbers, and market share metrics for UPI in recent years?

Evidence suggests that India's UPI payment system has experienced remarkable growth in transaction volumes, scaling from 0.02 billion transactions in 2017 to over 83 billion in 2024 [9][10].

Further demonstrating this strong upward trajectory, evidence suggests that in FY 2025-26, UPI processed 24,161.69 crore transactions with an impressive annual value of ₹314 lakh crore [3].

Reflecting its dominant position on the international stage, evidence suggests that India accounts for approximately 49% of global real-time payment transactions as of 2023 [2].

3.3 What are the primary economic and technological causes behind the rapid adoption of UPI across India?

Evidence suggests that the rapid adoption of India's UPI payment system has been significantly propelled by major macroeconomic events and supportive regulatory measures [10][8]. Specifically, the 2016 demonetization policy invalidated 86% of cash by

value, which forced an immediate shift toward digital alternatives [10]. Furthermore, evidence suggests that the COVID-19 pandemic subsequently doubled UPI transaction volumes in a single year [9][10].

On the technological and structural front, evidence suggests that the implementation of the Zero Merchant Discount Rate in 2019 eliminated transaction fees for merchants to prioritize adoption velocity [10]. This was complemented by integration with Aadhaar and eKYC protocols, which reduced merchant onboarding times from weeks to minutes [10].

Additional factors such as business ecosystem development, measured by formal UDYAM registrations, appear to be stronger drivers of UPI adoption than internet connectivity [9][10]. Evidence suggests that once a threshold of high-speed connectivity is reached, further infrastructure improvements have little impact on user onboarding [9]. Ultimately, the growth of digital payments driven by innovation and regulatory support has meant that UPI has successfully facilitated the growth of fintech firms, online shopping, and small businesses [6][8].

3.4 What effects has the widespread implementation of UPI had on traditional banking, cash usage, and financial inclusion in India?

The widespread implementation of the UPI payment system in India has driven significant transformations across the country's financial landscape, contributing directly to increased financial inclusion, reduced cash usage, and faster payment processing [2][6][7][3]. Because payments are recognized as an effective means to achieve financial inclusion by reaching all segments of society, the adoption of UPI serves a critical developmental role [8].

Evidence suggests that person-to-merchant transactions account for 63 percent of the total UPI transaction volume [3]. This high volume of merchant transactions highlights how deeply digital payments are integrated into daily commerce, aligning with the observed reduction in traditional cash usage and the broader push toward comprehensive financial inclusion [2][6][7][3].

3.5 What are the main disagreements, security concerns, or debates surrounding UPI transaction failures and market concentration risks?

No finding for this question was supported by two independent websites or by one tier 1 source, so nothing is reported here. See the appendix for the claims that could not be checked.

3.6 What recent developments and international expansions have occurred for India's UPI payment system?

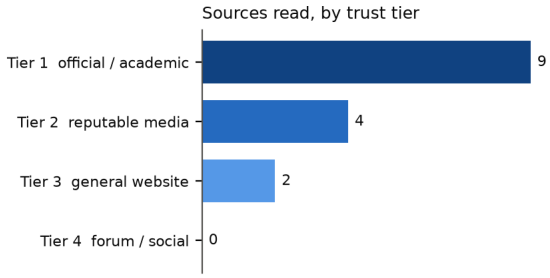
Evidence suggests that India's Unified Payments Interface (UPI) has been expanding internationally through strategic system linkages [8]. Specifically, India's UPI and Singapore's PayNow were successfully linked in February 2023 [8]. This development was designed to facilitate and streamline cross-border payments between the two nations [8].

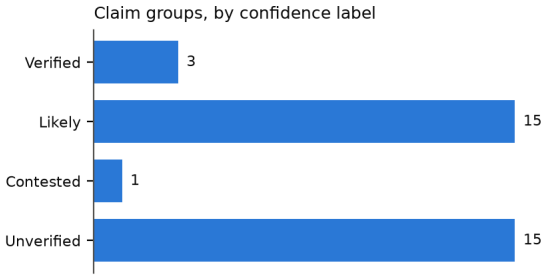
4. Contested points

- UPI is currently operational in 8 countries, including Singapore, France, and the UAE [3]. Disputed by: "UPI is currently operational in seven countries, including the UAE, Singapore, Bhutan, Nepal, Sri Lanka, France, and Mauritius." [2]; "UPI is currently

available for use across nine countries." [5]. The claims disagree on the exact number of countries where UPI is operational and mention different sets of countries or external implementations.

5. Source quality





Ref	Title	Website	Tier	Read
[1]	Unified Payments Interface - Wikipedia	wikipedia.org	2	full text
[2]	UPI: Revolutionizing Digital Payments in India	pib.gov.in	1	full text
[3]	UPI completes 10 glorious years, Emerges as World’s Largest Real-Time Payments Platform, Anchoring India’s Dig	pib.gov.in	1	full text
[4]	NPCI scales up UPI adoption in UAE to strengthen digital ties - The Economic Times	indiatimes.com	2	full text
[5]	NPCI and J.P. Morgan Payments Power Real-Time FX for UPI	jpmorgan.com	3	full text
[6]	A Secondary Data Analysis Approach to Evaluate the Impact of UPI on the Digital Transactions In India	doi.org	1	full text
[7]	UPI and financial inclusion in rural India: A case study	sciencedirect.com	1	abstract
[8]	Publications	rbi.org.in	1	full text
[9]	Supply side drivers of digital payment adoption in India	springer.com	1	full text
[10]	Supply side drivers of digital payment adoption in India - Discover Sustainability	springer.com	1	full text
[11]	South-East Asian markets on UPI radar as NPCI pushes global expansion	business-standard.com	2	full text
[12]	UPI market-share cap faces enforcement hurdle as PhonePe, Google Pay retain nearly 80%	thehindubusinessline.com	3	full text
[13]	UPI continues to lead as digital transactions soar 35% in FY25: RBI	indiatimes.com	2	full text
[14]	Dynamics of Globalization and Inclusive Economic Transformation in India: Interlinkages Between Financial Deve	doi.org	1	full text
[15]	Exploring the Impact of Debit Card Ownership on Financial Inclusion in ASEAN+6 Countries	doi.org	1	full text

6. Limitations

- Trust tiers and confidence labels are estimates based on the type of website and on agreement between sources. They do not prove that a statement is true, so important facts should be checked in the cited sources.

- 15 claims found in only one low-trust source were left out of the findings (see the appendix).
- 1 claim disputed by credible sources are shown in the contested points section.
- No checked findings were available for: Q5.
- 5 pages could not be opened (paywalls, blocked downloads or pages without text).
- For 1 scholarly source only the abstract was available, not the full paper.
- Sources are limited to what DuckDuckGo, Wikipedia and OpenAlex returned during this run, in English.

7. References

[1] **Unified Payments Interface - Wikipedia**. *wikipedia.org*, 2016-04-12. Tier 2, weight 0.65. https://en.wikipedia.org/wiki/Unified_Payments_Interface (accessed 10 October 2026)

[2] Saurabh Kalia. **UPI: Revolutionizing Digital Payments in India**. *pib.gov.in*. Tier 1, weight 0.95. https://static.pib.gov.in/WriteReadData/specificdocs/documents/2024/dec/doc2024121462101.pdf (accessed 10 October 2026)

[3] **UPI completes 10 glorious years, Emerges as World's Largest Real-Time Payments Platform, Anchoring India's Digital Economy**. *pib.gov.in*, 2025-11-20. Tier 1, weight 0.95. https://pib.gov.in/PressReleaseIframePage.aspx?PRID=2257087 (accessed 10 October 2026)

[4] PTI Follow us Share Font Size Abc Small Abc Medium Abc Large Save Print Comment. **NPCI scales up UPI adoption in UAE to strengthen digital ties - The Economic Times**. *indiatimes.com*, 2026-10-10. Tier 2, weight 0.8. https://economictimes.indiatimes.com/nri/invest/npci-scales-up-upi-adoption-in-uae-to-streng... (accessed 10 October 2026)

[5] **NPCI and J.P. Morgan Payments Power Real-Time FX for UPI**. *jpmorgan.com*, 2026-06-30. Tier 3, weight 0.45. https://www.jpmorgan.com/payments/newsroom/npci-cross-border-upi-real-time-fx (accessed 10 October 2026)

[6] Kashish Talwani; JAAFR editor cheif. **A Secondary Data Analysis Approach to Evaluate the Impact of UPI on the Digital Transactions In India**. *doi.org*, 2020-01-01. Tier 1, weight 1.0. https://doi.org/10.56975/jaafr.v4i4.508512 (accessed 10 October 2026)

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[10] Motheram; Abhinav; Buteau; Sharon. **Supply side drivers of digital payment adoption in India - Discover Sustainability**. *springer.com*, 2026-04-10. Tier 1, weight 1.0. https://link.springer.com/article/10.1007/s43621-026-02901-x (accessed 10 October 2026)

[11] Business Standard; Ajinkya Kawale. **South-East Asian markets on UPI radar as NPCI pushes global expansion**. *business-standard.com*, 2026-06-01. Tier 2, weight 0.8. https://www.business-standard.com/economy/news/south-east-asian-markets-on-upi-radar-as-npci... (accessed 10 October 2026)

[12] Jyoti Banthia. **UPI market-share cap faces enforcement hurdle as PhonePe, Google Pay retain nearly 80%**. *thehindubusinessline.com*, 2026-09-28. Tier 3, weight 0.5. https://www.thehindubusinessline.com/money-and-banking/upi-market-share-cap-faces-enforcemen... (accessed 10 October 2026)

[13] Www Etbfsi Com; ETBFSI Research. **UPI continues to lead as digital transactions soar 35% in FY25: RBI**. *indiatimes.com*, 2025-05-29. Tier 2, weight 0.8. https://bfsi.economictimes.indiatimes.com/articles/record-growth-in-digital-transactions-upi... (accessed 10 October 2026)

[14] Dr T SUBRAMANI. **Dynamics of Globalization and Inclusive Economic Transformation in India: Interlinkages Between Financial Development, Energy Consumption, Employment, Poverty, and Sustainable Economic Growth**. *doi.org*, 2026-05-17. Tier 1, weight 1.0. https://doi.org/10.55041/ijcope.v2i5.533 (accessed 10 October 2026)

[15] Yesi Hendriani Supartoyo; Latif Adam; Erla Mychelisda; Md Asadul Islam. **Exploring the Impact of Debit Card Ownership on Financial Inclusion in ASEAN+6 Countries**. *doi.org*, 2026-07-29. Tier 1, weight 1.0. https://doi.org/10.21512/jas.v13i2.12653 (accessed 10 October 2026)

Appendix A. Claim verification table

Every group of matching claims found by the Verifier, with the sources that support or dispute it.

Claim	Label	Supported by	Disputed by	Sites
Q1 The Unified Payments Interface (UPI) was launched on 11 April 2016 by the National Payments Corporation of India (NPCI) under the regulatory oversight of the Reserve Bank of India (RBI).	VERIFIED	[1] [2] [3]	-	2
Q1 UPI is a real-time, account-to-account payment system that enables instant, secure transactions via mobile apps.	VERIFIED	[4] [5]	-	2
Q1 UPI consolidates multiple bank accounts into a single mobile interface.	LIKELY	[8]	-	1
Q1 UPI is fully interoperable, allowing any app to pay any QR code.	UNVERIFIED	[12]	-	1
Q2 UPI transaction volumes grew from 0.02 billion in 2017 to over 83 billion in 2024.	LIKELY	[9] [10]	-	1
Q2 In FY 2025-26, UPI processed 24,161.69 crore transactions with an annual value of ₹314 lakh crore.	LIKELY	[3]	-	1
Q2 India accounts for approximately 49% of global real-time payment transactions as of 2023.	LIKELY	[2]	-	1
Q2 As of 2025, UPI accounts for 84% of digital payments in India.	UNVERIFIED	[1]	-	1
Q2 UPI handles more daily transactions than Visa.	UNVERIFIED	[1]	-	1
Q3 The 2016 demonetization policy, which invalidated 86% of cash by value, forced a shift toward digital alternatives.	LIKELY	[10]	-	1
Q3 The COVID-19 pandemic doubled UPI transaction volumes in a single year.	LIKELY	[9] [10]	-	1
Q3 The implementation of the Zero Merchant Discount Rate (MDR) in 2019 eliminated transaction fees for merchants to prioritize adoption velocity.	LIKELY	[10]	-	1
Q3 Integration with Aadhaar and eKYC protocols reduced merchant onboarding times from weeks to minutes.	LIKELY	[10]	-	1
Q3 Business ecosystem development, measured by formal UDYAM registrations, is a stronger driver of UPI adoption than internet connectivity.	LIKELY	[9] [10]	-	1
Q3 UPI has facilitated the growth of fintech firms, online shopping, and small businesses.	LIKELY	[6]	-	1
Q3 The growth of digital payments in India is driven by innovation and regulatory support.	LIKELY	[8]	-	1
Q3 Once a threshold of high-speed connectivity is reached, further infrastructure improvements have little impact on user onboarding.	LIKELY	[9]	-	1
Q3 The number of UPI QR codes in India increased by 91.5% to 65.8 crore.	UNVERIFIED	[13]	-	1
Q3 Leading UPI apps maintain their market position due to trust, merchant distribution, and economics.	UNVERIFIED	[12]	-	1

Claim	Label	Supported by	Disputed by	Sites
Q3 The absence of Merchant Discount Rate (MDR) since 2020 has resulted in payments generating little direct revenue for apps.	UNVERIFIED	[12]	-	1
Q3 Merchant payments above 2,000 rupees will generate revenue starting October 15, 2026.	UNVERIFIED	[12]	-	1
Q4 UPI has contributed to increased financial inclusion, reduced cash usage, and faster payment processing in India.	VERIFIED	[2] [6] [7] [3]	-	3
Q4 Person-to-merchant (P2M) transactions account for 63% of total UPI transaction volume.	LIKELY	[3]	-	1
Q4 Payments are recognized as a means to achieve financial inclusion by reaching all segments of society.	LIKELY	[8]	-	1
Q4 Retail digital transactions represented 99.9% of all non-cash retail payments in FY25.	UNVERIFIED	[13]	-	1
Q4 Debit card transactions in India declined for the second consecutive year.	UNVERIFIED	[13]	-	1
Q4 Dubai aims to reach 90 percent digital transactions by the year 2026.	UNVERIFIED	[4]	-	1
Q5 The government has set a deadline of December 31, 2026, for the 30 percent market-share cap on UPI apps.	UNVERIFIED	[12]	-	1
Q6 India's UPI and Singapore's PayNow were linked in February 2023 to facilitate cross-border payments.	LIKELY	[8]	-	1
Q6 NPCI International Payments is exploring bilateral UPI linkages with Indonesia, Thailand, and Malaysia.	UNVERIFIED	[11]	-	1
Q6 Over seven million Indian citizens travel to the UAE each year.	UNVERIFIED	[4]	-	1
Q6 NIPL is expanding UPI acceptance in the UAE through partnerships with financial institutions like NeoPay, Network International, and Magnati.	UNVERIFIED	[4]	-	1
Q6 J.P. Morgan Payments and NPCI are collaborating to enable real-time currency conversion and settlement for cross-border UPI transactions.	UNVERIFIED	[5]	-	1
Q6 UPI is currently operational in 8 countries, including Singapore, France, and the UAE.	CONTESTED	[3]	[2] [5]	1